

WHAT SHOULD DATA TEACH? MOVING BOTTLENECKS ACROSS CIRCUIT, STORE, AND USE

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ABSTRACT

What should data teach a language model at a particular point in training? A circuit view reveals three distinct bottlenecks: forming a computation, making its required content available, and selecting among available routes. A shared diagnosis-to-data principle connects them: localize the missing operation, preserve its causal relation, vary shortcut-bearing context, and re-audit the residual. Formation-sensitive selection and prerequisite ordering accelerate a binding–matching–transport path; a brief early prefix from the same training multiset retains a validation advantage through 100B tokens. Availability counterfactuals then distinguish writing content from invoking available memory, while paired supervision and context-opportunity ranking improve matched route decisions and long-context answer likelihood. A continuous 350M-model experiment connects the three interventions on the same facts: early circuit training improves subsequent learning, and the complete sequence outperforms stage-replacement controls on facts withheld from Use teaching. Independent query surfaces and opposed-source decisions expose conditional arbitration as the remaining frontier. Together, these results show why a change in the limiting operation calls for a change in supervision, not merely a new ranking of difficult examples.

1 INTRODUCTION

What limits a language model at a particular checkpoint? We began by asking whether the required computation had formed. Formation-time circuit signals organized an early prefix that accelerated a binding–matching–transport path. In the auxiliary order study, the same-multiset intervention improved Overall, Math, Code, and a natural induction-sensitive bucket, but not Web. On frozen Web cases, supplying the relevant evidence as context QA substantially improves target accuracy over source-masked and shuffled controls in both arms (Figure 4). The read path can use accessible evidence, raising the next diagnostic question: can model state supply the content the task requires?

This led from Circuit to Store. Availability counterfactuals separated targets that parametric memory could not supply from those available in isolation but suppressed in a natural carrier: writing and invoking memory were different operations. The latter exposed Use, where multiple information or computation routes are available and the query must determine which one controls the answer. Controlled supervision strengthens matched route decisions, but general arbitration across independently phrased and opposed routes remains difficult.

Our central claim is that the useful training target changes as a limiting operation is repaired: computation formation, content availability, and control among available routes call for different supervision. Circuit, Store, and Use are operational regimes rather than physical modules or a universal training clock. Natural-text residuals test what remains after formation; available-source conflicts test what remains once content can be supplied.

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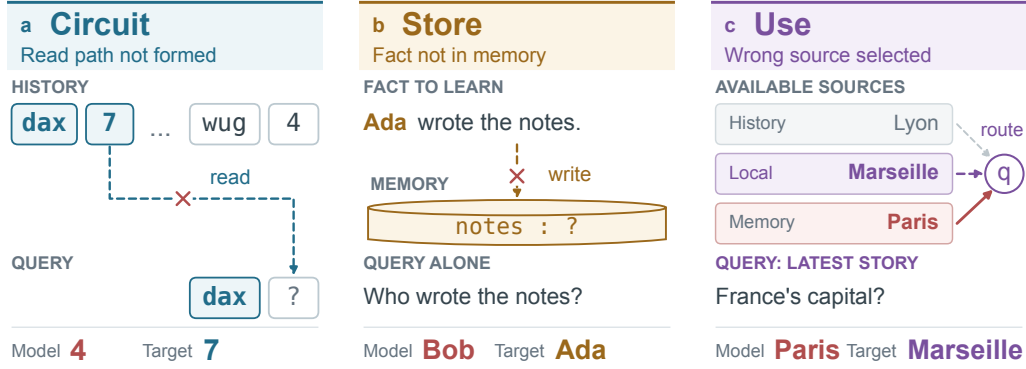


Figure 1: **Three failures, three teaching targets.** (a) Repeated dax should retrieve 7; the read fails. (b) “Ada wrote the notes” has not been acquired for unaided recall. (c) Query q asks for the latest fictional story’s capital, Marseille, but memory supplies Paris. Dashed paths denote candidate operations; crosses mark failed read/write steps; the solid red arrow marks wrong routing. Bottom entries compare model predictions (red) with target answers (panel colors). These are illustrative cases, not measured outputs.

One diagnosis-to-data principle connects the chain: localize the missing operation, preserve its defining relation, vary shortcut-bearing context, and re-audit the remainder. We first isolate the demands and their repairs, then connect them in a continuous training experiment on the same facts. Stage-replacement controls test whether early circuit organization, target-fact writing, and paired Use supervision contribute to one final task (Section 7).

Figure 1 illustrates the three failures. H denotes remote history accessed through an induction-style read, L nearby context, and M parametric memory.

Our contributions are:

- **A circuit-level reinterpretation of training bottlenecks.** Starting from Circuit formation, we show that residual failure can move from a missing computation to unavailable content and then to incorrect control among available routes. Circuit, Store, and Use are therefore operationally distinct failures rather than interchangeable notions of example difficulty.
- **A shared diagnosis-to-data principle with different repair actions.** Frozen failures, availability counterfactuals, and mechanism-sensitive signals identify the missing operation; relation preservation and context replacement then construct supervision suited to Circuit formation, Store acquisition or invocation, and matched Use decisions.
- **A discovery chain with sequential validation.** Early Circuit organization retains an advantage through 100B tokens. A separate continuous experiment connects Circuit, Store, and Use on common facts and improves final use over stage-replacement controls. Independent-query and opposed-source audits identify transferable conditional arbitration as the next bottleneck.

2 RELATED WORK

Curriculum and adaptive selection. Curriculum and model-aware selection organize examples by difficulty, quality, prerequisites, or checkpoint-local influence (Chen et al., 2023a; Wettig et al., 2024; Zhang et al., 2026; 2025; Yu et al., 2024; 2025; Lee et al., 2026a; Wang et al., 2025; Schoenegger et al., 2025). Our distinction is not that data should change with training, but *what supervision must change*: computation formation, content availability, and route selection require different repair actions rather than only a new ranking score.

Mechanistic data and circuit formation. MDA traces examples to interpretable units and accelerates induction-head convergence across Pythia scales (Chen et al., 2026); Bi-Induct separates

induction signatures from functional ICL gains (Sabry & Belz, 2026). SMDA attributes training influence to symbolic behavioral policies (Habibi et al., 2026), while SAMS schedules circuit-steered data by training utility (Lee et al., 2026b). Our distinction is the diagnosis of missing computation, unavailable content, and incorrect route control, and the different repair actions these require. MDA supplies the Circuit ranking signal; we test formation timing, prerequisites, exit, and persistence through 100B, then connect the repairs in a common training chain. Induction work motivates binding–matching–transport (Olsson et al., 2022; Singh et al., 2024); shortcut results motivate varied realizations (Kawata et al., 2025).

Storage and incomplete learning. Knowledge exposure, extraction, and manipulation are distinct problems (Kandpal et al., 2023; Allen-Zhu & Li, 2024; 2025; Zhou & Zhang, 2025; Xue et al., 2026). Our Store category is functional: counterfactuals separate content that M-only cannot supply from content that M supplies but that loses control in a natural carrier, without asserting one physical memory location (Geva et al., 2021).

Long context and route control. Context–memory conflicts reveal task-dependent source reliance (Kortukov et al., 2024; Xu et al., 2024; Sun et al., 2026), while long-context selectors use attention or long–short loss contrasts to find informative windows (Chen et al., 2025; Wu et al., 2025; Fang et al., 2025; Deng et al., 2026). Our clipped full–local response targets context opportunity; unlike confidence-weighted long–short selectors, its role here is a matched test of window identity. We distinguish that predictive dependence from learning which route a query authorizes, and separate matched-route and ranking gains from general arbitration across opposed routes.

3 A CIRCUIT VIEW OF TRAINING

3.1 FROM RESIDUAL FAILURE TO MISSING OPERATION

Let $u = (x, y, q, S)$ denote context, target, query, and candidate sources. At checkpoint θ_t , source-retain/remove/swap/truncation views assign a behavioral failure class $f_t(u)$. Separately, component sensitivity G_c measures a gradient response, while M_k aggregates the classified residual,

$$G_c(u, t) = \frac{\|\nabla_{\theta_c} \ell_t(u)\|_2^2}{|\theta_c|}, \quad M_k(t) = \sum_u \mathbf{1}[f_t(u) = k] \max\{\text{NLL}_t(u) - 2, 0\}, \quad (1)$$

where M_k is excess-loss mass on fixed identities. Counterfactual gates determine f_t , not the magnitude of G_c . Gradients prioritize components within the gated population; controlled binding, matching, transport, ablation, and swap tests assess function. MDA subsequently ranks training examples for an identified Circuit role.

CIRCUIT has an unambiguous source but no usable read path. STORE lacks required M-only content. USE has available routes but the task-authorized one fails to control output. The Store study tests this boundary: once M-only succeeds, carrier suppression is a Use failure. Local denotes residual explained by nearby context in the natural-case audit; it is a descriptive category, distinct from arbitration between competing sources. Gates and precedence appear in Appendices B and G.

3.2 FROM DIAGNOSIS TO DATA

Separate an operation schema r from a context realization z :

$$u = (r, z), \quad y = \pi_r(z), \quad \mathcal{D}_r^Q = \{(r, z_i, \pi_r(z_i)) : z_i \sim Q(\cdot | r)\}_{i=1}^n. \quad (2)$$

Construction holds schema r fixed while replacing realization z : a role-complete read schema, a cue–value relation, or a query-authority rule. $P(\cdot | r)$ denotes task-valid contexts; Q is the chosen training distribution with support within P . Each answer follows π_r .

Selection scores enter only after this operation gate. Formation-time MDA is one ranking signal for a missing Circuit role, whereas the full–local response ranks windows where remote context can change prediction. Their teaching meaning comes from the gated operation and relation-preserving controls, not score magnitude alone.

Concentration raises the density of r ; variation prevents token, carrier, domain, or source identity from naming the answer without performing it. Table 1 shows how the principle yields different actions and residuals.

Table 1: **The residual-bottleneck chain in our investigation.** Each intervention addresses an identified missing operation; the remaining bottleneck motivates the next stage.

Stage	Missing operation	Diagnostic	Intervention	Residual bottleneck →
CIRCUIT	path not formed	component gradients / formation-time MDA	mechanism-matched ordering	content unavailable → Store
STORE	required content unavailable	M-only / evidence counterfactuals	Write; test invocation at the boundary	selection conflict → Use
USE	which circuit/source wins	query-conditioned routing probes	paired routing / CCUR	multi-arm arbitration → open

The primary 1.485B model runs through 100B tokens; auxiliary 160M studies isolate order. Store and Use studies start from the completed 100B model. A separate 350M chain applies the three interventions sequentially to a common factual task (Section 7). Appendix A identifies each model and comparison.

3.3 DECISION-CHANGING VARIATION

For a feasible context block j (grouping variables that must change together), define

$$j \in \mathcal{J}_r \iff \exists z, z' \in \text{supp } P(\cdot | r) : z_{-j} = z'_{-j} \wedge \pi_r(z) \neq \pi_r(z'). \quad (3)$$

Here z_{-j} fixes the remaining blocks. In an opposed-source pair, hold the fact, passage, and candidate values fixed; change the task-authority instruction as one block. If this switches the required answer, that block belongs to $\mathcal{J}_r$. Equivalent paraphrases should instead preserve the answer. Equation 3 specifies feasible construction contrasts, not an estimator of a latent language distribution.

Circuit varies tokens and domains around a role graph; Store varies cues around an assignment and audits wrong-pair training. Use must cover opposed authority decisions as well as equivalent wordings: rewarding one source throughout permits a marginal preference instead of the conditional rule.

3.4 HOW THE BOTTLENECK MOVED IN OUR INVESTIGATION

On 272 fixed natural identities, checkpoint-local diagnosis shows earlier Circuit contraction under Curriculum (Figure 2a). A separate available-source conflict panel retains a Use-heavy residual after formation (b). These complementary audits motivate the next repair; Section 7 then tests the interventions on one common training chain and factual panel.

4 CIRCUIT: FORMING THE COMPUTATION

Localization. Information is present, but the required computation has not formed. Controlled probes separate previous-token binding, content matching, and value transport. Binding becomes usable first, while top-ranked head identities change during formation; the invariant is the role-complete relation, not a head coordinate. Formation-time signals rank natural examples. Natural-text ambiguous recall—a repeated predecessor–target pair whose predecessor has multiple observed continuations in the available document prefix—tests transfer beyond the synthetic construct.

Formation intervention. **Fixed:** the role-complete binding–matching–transport schema. **Varied:** natural domains, sources, token identities, and lexical realizations. **Audit:** functional formation, teacher time, order, exit, and ordinary continuation. Let $\mathcal{C}_{\text{match}}$ be the Q/K component set frozen by the held-out matching gate and $\mathcal{D}_d$ the candidate pool in domain d . The formation teacher is the within-domain selection

$$\mathcal{S}_{\text{match}}(\tau, d) = \text{TopK}_{z \in \mathcal{D}_d} \text{MDA}_{\mathcal{C}_{\text{match}}}(z; \theta_\tau), \quad (4)$$

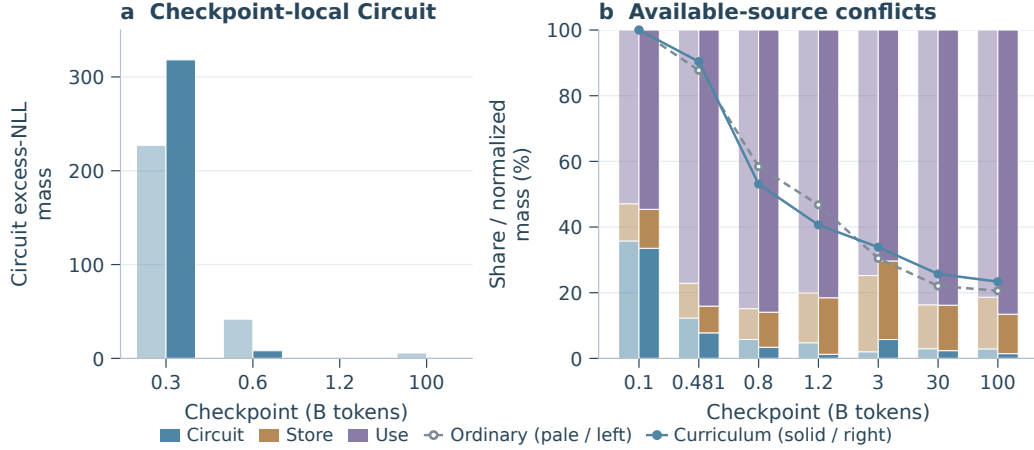


Figure 2: **Circuit residual contracts; available-source conflicts persist.** (a) Circuit excess-NLL mass from checkpoint-local diagnosis on 272 fixed natural identities. (b) Composition of a separate 942-target conflict residual; lines normalize total mass to each arm’s 0.1B value. Pale/left bars: Ordinary; solid/right: Curriculum. Appendix B gives roster construction, rules, and sensitivity.

where τ is the formation or endpoint checkpoint and the slot budget is fixed. MDA supplies the component-sensitive score; our intervention adds role decomposition, prerequisite order, and formation-guided exit. Joint-to-matching exposes the relation before concentrating matching demand (Appendix Figure 7). A pilot identifies the formation window and selects the exit by held-out performance; this schedule is frozen before the 100B run.

Formation evidence. Curriculum advances matching while Ordinary still shows little matching selectivity, although binding is usable in both arms (Figure 3a). With matching-only organization fixed, formation-time MDA improves Overall and ambiguous recall prediction, with lower endpoint NLL in every domain (Figure 3b,c). This isolates teacher timing: the endpoint identifies a functioning path, whereas formation-time examples accelerate its construction. With membership fixed, Joint-to-matching beats interleaving and reverse order (Figure 4a; Appendix Table 15); the exit sweep peaks after the measured path forms (Table 16). These controls separate organization from selection and calibrate exit around functional formation.

Persistent pretraining gains. Against matched interleaving, the same-multiset Joint-to-matching order improves Overall, Math, and Code PPL but not Web (Figure 4a; Appendix Table 15). Separately, Curriculum improves Overall PPL over Ordinary across three independent 3B runs, and both auxiliary runs pass the frozen binding, matching, and transport gates (Tables 18 and 12). In the target-scale trajectory, Curriculum surpasses Ordinary’s final validation performance at 82B rather than 100B tokens, then retains its advantage through the endpoint (Figure 3d). Every endpoint domain favors Curriculum (Table 19). The intervention uses a selected 1.2B-token prefix within the same total training multiset, followed by the ordinary sampling schedule.

MDA establishes that mechanistic augmentation accelerates circuit convergence (Chen et al., 2026). We extend this to long-horizon general pretraining: to our knowledge, this is the first evidence that a circuit-sensitive early prefix retains a validation advantage after the intervention ends and ordinary training continues to 100B tokens.

Residual bottleneck. Despite the Web exception above, both arms answer the same frozen Web facts substantially better with relevant context than with a masked or shuffled source (Figure 4b). The read path is usable, motivating the next test: can parametric memory supply the required content? The separate 350M chain further shows that early Circuit training improves subsequent memory acquisition (Appendix H, Table 41), connecting read-path formation to the next intervention.

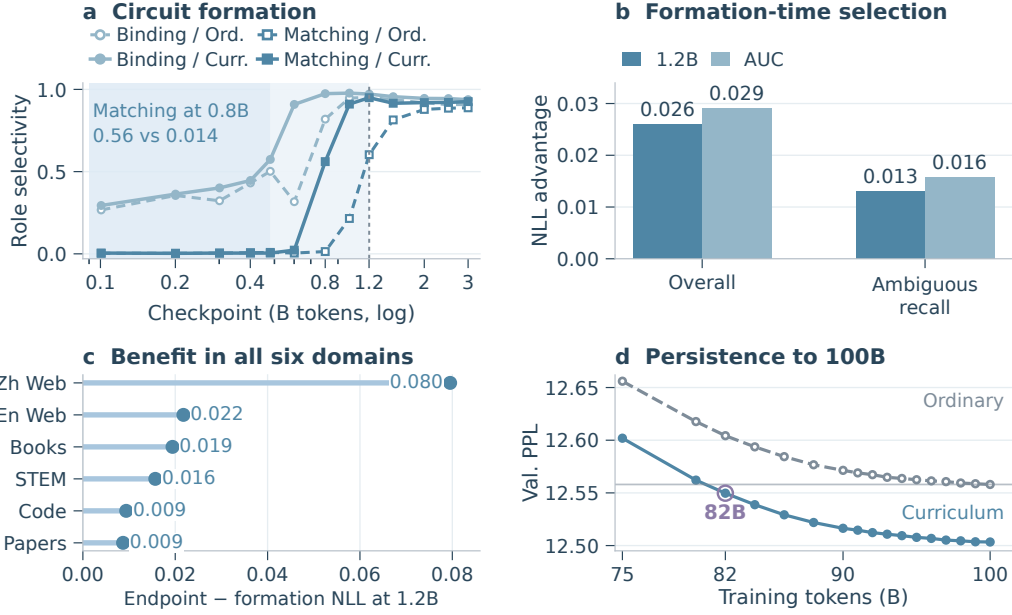


Figure 3: **Circuit training advances formation with persistent gains.** (a) Binding forms in both arms; reordering advances matching. (b) Endpoint-minus-formation NLL is positive for Overall and ambiguous recall at 1.2B and over normalized 0.1–1.2B AUC; 1.2B PPL reductions are 2.57% and 1.30%. (c) The same strict matching-only teacher comparison favors formation-time selection in all six domains at 1.2B. (d) Curriculum crosses Ordinary’s final PPL at 82B and stays lower through 100B. Appendix C gives teacher trajectories, order controls, and the measured-exit sweep.

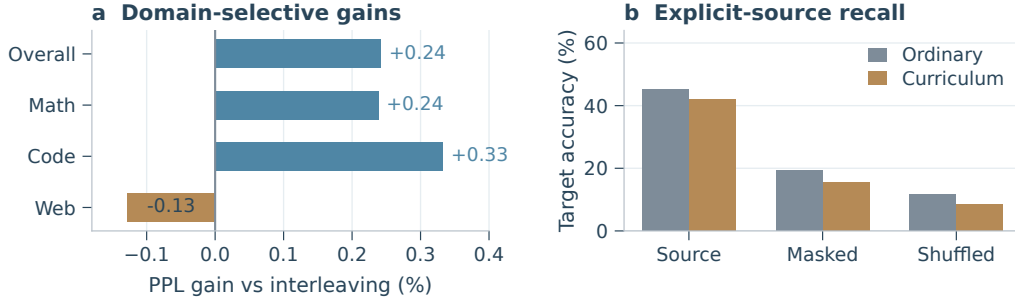


Figure 4: **Domain-selective gains motivate an availability audit.** (a) At 2.5B in the 160M order study, fixed-multiset Joint-to-matching improves Overall, Math, and Code PPL over interleaving, but not Web. (b) Relevant-source, masked-source, and shuffled-context QA on 435 fixed Web facts at 2.5B shows both arms use explicit evidence. Appendix Tables 11 and 15 give full presentation/NLL comparisons and reverse-order controls, respectively.

5 STORE: MAKING CONTENT AVAILABLE

Localization. The Web residual left by Circuit training raises a different question: given a usable computation, is the content it needs available in model state? We use Store as a functional availability category, not as a claim that one architectural module physically contains memory. Knowledge exposure, availability, and invocation are not equivalent. M-WRITE targets cases where M-only cannot supply the value; M-USE targets cases where M-only succeeds but the value loses control under a natural carrier. Let $m_\theta(u; x)$ be the correct-minus-competing answer log-probability margin at the first divergent answer token. For explicit-evidence, M-only, and natural-carrier views, define evidence rescue and circuit drag,

$$\Delta_E(u) = m_\theta(u; x_E) - m_\theta(u; x_M), \quad D_{\text{drag}}(u) = m_\theta(u; x_M) - \bar{m}_\theta(u; x_F). \quad (5)$$

Here $\overline{m}_\theta$ averages the two natural carriers. The natural availability gate requires M-only candidate failure, evidence-present success, and sufficient evidence rescue. An invocation case instead requires M-only success and suppression under both answer-free carriers. These views separate missing content from suppressed content and generic difficulty (Appendix Table 20; protocols in Appendix D.2).

Repair intervention. **Fixed:** cue–value relation and answer. **Varied:** cue wording, carriers, and optimizer positions. **Audit:** relation direction, unseen cues, and natural invocation. M-WRITE distributes independently supervised cue–value segments across updates. Wrong-pair training preserves answer-token exposure but changes the assignment. When M-only succeeds, M-USE places the value behind the suppressing natural carrier. The same fact therefore calls for different supervision depending on availability (Appendix Figure 8).

Relation acquisition. On controlled novel relations, the first test asks whether M-WRITE learns the assignment rather than an answer-token prior. A low writing dose shifts the answer margin toward the intended value under both exact and unseen cues; wrong-pair training shifts it toward the swapped assignment (Appendix Tables 22 and 23). This assignment-sensitive response across cue forms demonstrates directional relation learning. We next test which supervision helps natural facts that are already available.

Acquisition versus invocation. The second test freezes 512 available but circuit-dragged facts and crosses two intervention slots; disabling an operation substitutes matched control facts rather than removing tokens. Use contributes more natural-carrier improvement than Write, with only a small interaction (Table 2; Appendix Table 25). This difference also appears in candidate accuracy under alternate prompt B, which Use training does not see. Write raises M-only gain at either Use setting. Thus the operative distinction is what supervision helps available content control a prediction, not simply whether the model has encountered the fact.

Table 2: **Available facts benefit most from invocation training.** Parent-minus-continuation NLL gains ($\uparrow$); B accuracy is first-divergent-token candidate choice, not free generation. All arms share 512 facts, two matched slots, parent, and 25M budget. Use trains carrier A; natural gain averages A/B on those facts. Appendix D gives carrier and factorial contrasts.

Arm	Active operation	M-only gain	Natural-use gain	B accuracy (%)
C0	matched controls	0.636	0.894	82.03
C1	Write	0.740	0.971	83.98
C2	Use	0.723	1.201	93.36
C3	Write + Use	0.806	1.294	95.31

Residual bottleneck. In the common-fact chain, Store lowers memory NLL, and replacing target writing with control facts weakens final joint use (Appendix H, Tables 40 and 42). Store separates values that must be acquired from those that are available but fail to control prediction. The latter is no longer missing content; it is control over available information and computation routes. We therefore turn to Use: which route should the task authorize?

6 USE: SELECTING AMONG AVAILABLE ROUTES

Localization. Once computation and content are available, Use asks which route should control the answer. History (H), Local context (L), and parametric Memory (M) may offer competing values. The residual in Figure 2b motivates teaching selection under the query and authority relation.

Application intervention. **Fixed:** the task-authorized winner rule or full–local opportunity criterion. **Varied:** query surface, facts, documents, sources, and candidates. **Audit:** independent surfaces, disjoint documents, and opposed routes. The invariant is not “prefer H/L/M”: changing authority must change the winner when the sources disagree (Appendix Figure 9).

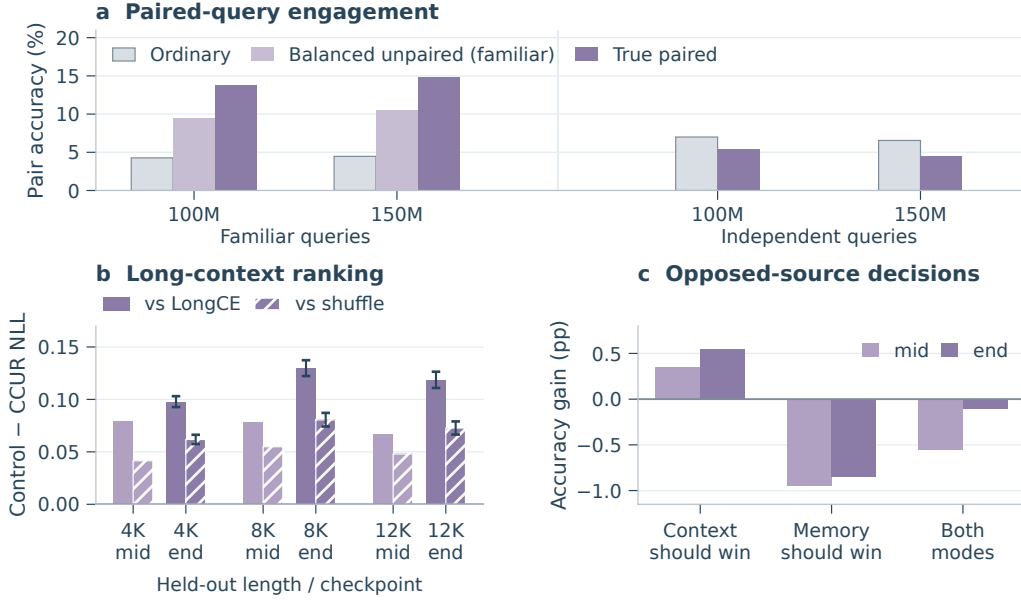


Figure 5: **Matched supervision improves engagement and long-context ranking.** (a) Paired supervision outperforms Ordinary and balanced unpaired on familiar queries; independent phrasings test transfer. (b) On a document-disjoint training pool, CCUR beats LongCE and score-shuffle; whiskers are paired 95% endpoint intervals. (c) CCUR minus LongCE accuracy on opposed queries shows a source-prior shift, not joint arbitration. Positive values in (b,c) favor CCUR.

U1: matched-route engagement. True pairing teaches both authority modes per fact; balanced unpaired teaching assigns one mode per fact, balanced across facts. Both use ordinary language modeling plus the same auxiliary target CE and competitor-margin loss. True pairs share one per-fact template; unpaired decisions use the other two templates, so the comparison tests the complete constructions (Appendix Table 26). On unseen fact families with familiar queries, paired supervision improves strict-pair accuracy by 4.4 percentage points over balanced unpaired teaching at +100M (Figure 5a). On Store-taught facts withheld from Use teaching, the continuous chain likewise improves joint use after paired supervision and outperforms the unpaired replacement (Figure 6; Table 3; Appendix H). Reversed supervision reverses an independent route offset (Table 33), supporting relation engagement under matched query surfaces.

U2: task-aligned long-window ranking. For scored positions I_x in a 16K window, Counterfactual Context-Use Ranking (CCUR) measures the clipped full-local response,

$$s_{\text{CCUR}}(x) = \frac{1}{|I_x|} \sum_{i \in I_x} \text{clip}(\min\{\ell_i^L, \ell_i^R\} - \ell_i^F, 0, \log 5), \quad (6)$$

where ℓ_i^F is full-window NLL; ℓ_i^L, ℓ_i^R use the same local tokens with reset/original positions (Appendix E). CCUR and matched LongCE (Fang et al., 2025) share objective, source partition, quotas, updates, budget, and difficulty strata; only window identity changes within each cell. A score shuffle preserves the within-stratum score multiset but breaks its association with window identity. On a document-disjoint pool, CCUR lowers RULER-style answer-token NLL relative to both controls at every evaluated length and checkpoint, with paired endpoint intervals excluding zero throughout (Figure 5b; Appendix Table 30).

Conditional arbitration as the next bottleneck. Matched relation engagement does not reliably transfer to independently phrased queries: expanding the training families still leaves strict-pair accuracy below Ordinary at both saved checkpoints (Appendix Table 28). CCUR likewise trades improved context-authorized decisions for worse memory-authorized decisions, without improving joint success (Figure 5c; Table 32). These audits distinguish strengthening a route from learning

when it should win: the remaining target is a query-conditioned decision that transfers across wording and reverses with authority.

Section 7 assembles the stages into one continuous training chain; Appendix E gives the full transfer audits.

7 CONNECTING THE STAGES

Do the three repairs contribute to one final task? Two 350M routes share initialization, the 3B pretraining multiset, and domain slots; only the early order differs. Each continues through 50M Store and 100M Use, inheriting its parent state. We track the same 640 facts, taught during Store but excluded from Use teaching, under fixed reading, memory, and opposed-authority queries. Appendix H gives the branches and complete outcomes.

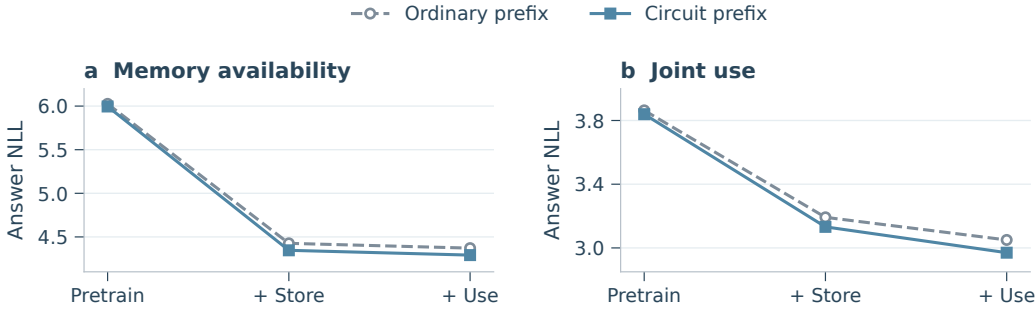


Figure 6: **The repairs compose on common facts.** Both routes receive target Store and paired Use after pretraining. On 640 Store-taught, Use-untaught facts, memory improves mainly during Store; the joint-use task improves further during Use. Each point evaluates the same fixed queries; lower answer-token NLL is better.

The early-course advantage extends into subsequent learning (Figure 6): Store reduces memory NLL more under the curriculum route, and Use produces a larger additional joint-task gain (Appendix Table 41). At endpoint, the complete sequence outperforms all four controls on joint-use NLL and strict-pair candidate accuracy (Table 3). Replacing target writing or paired teaching weakens final use despite retaining the other stages, connecting all three interventions through one common outcome.

Table 3: **Each stage contributes to final use.** Same 640 facts and 1,920 familiar-query pairs. Pair accuracy requires both authorized candidates to win at their first divergent token. “Control facts” replaces target Store; “unpaired” replaces the complete Use construction.

Training route	Joint-use NLL ↓	Pair accuracy (%) ↑
Complete sequence	2.970	5.26
No early curriculum	3.050	3.39
Control facts in Store	3.245	1.41
Unpaired Use	3.019	3.18
Ordinary continuation	3.870	0.99

The worked case in Appendix Table 43 makes the handoff concrete: the model can read Newton’s death year before recalling it unaided; Store supplies recall, and Use subsequently resolves the illustrated opposed-query pair. The aggregate gain is on familiar instructions; the cross-surface arbitration frontier identified above remains.

8 CONCLUSION

Circuit formation, content availability, and route control require different supervision. Our diagnostic studies identify these demands; a continuous common-fact experiment shows how their interventions compose, while the early Circuit advantage persists through 100B tokens in the long-horizon study. The remaining challenge is conditional arbitration that transfers across wording and opposed winners. The same principle guides that next step: preserve the required operation while varying the context that could substitute a shortcut.

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A SHARED SETUP AND SCOPE

A.1 PRIMARY SCALE RUN

Table 4: Configuration of the matched 1.485B-parameter/100B-token experiment.

Block	Item	Specification	Pair status
Model	Architecture	GPT decoder; 28 layers; $d = 2,048$; SwiGLU	matched
Model	Parameters	1,484,900,352	matched
Model	Attention	16-head MHA; RoPE; SDPA	matched
Model	Norm / vocab	RMSNorm; 65,536 tokens	matched
Input	Sequence / batch	2,048 tokens; 512 sequences (1,048,576 targets/update)	matched
Train	Optimizer	AdamW; $\beta = (0.9, 0.95)$; weight decay 0.1	matched
Train	Learning rate	2×10^{-4} peak; 2×10^{-5} minimum	matched
Train	Schedule	approximately 300M warm-up; cosine decay to 100B	matched
Data	Fixed	initialization, multiset, domain slots, optimizer, LR, budget	matched
Data	Changed	positions of the same examples	only change

The Curriculum prefix places joint binding–matching examples before matching–selected examples; Ordinary encounters the same sequences elsewhere in the fixed multiset. Domain weights and validation targets are identical across arms. Table 35 records the prefix allocation and data sizes. The long-horizon experiment contains one training trajectory per arm; the early comparison uses three independent initializations (Table 18).

Table 5: Evidence map. Separate rows use separate populations; shared terminology does not merge them.

Evidence	Model / start	Matched object	Support
Moving residual	1.485B pair	272 dynamic identities; 32 fixed-stratum cases; 942 conflicts	separate frozen panels
MDA teacher time	350M proxy; 1.485B target	formation-time vs endpoint-time MDA; matching-only selection	1 run per arm
Circuit confirmation	Pythia-style 160M @100M tokens	clean/swap, binding ablation, forced transport	2 runs $\times$ 3 domains
Order	Pythia-style 160M	same selected multiset; 2.5B endpoint	1 run per arm
Exit	1.485B target	same multiset; 3B endpoint	1 run per arm
Long horizon	1.485B target	same 100B multiset, positions only	dense 75–100B trajectory
M availability	Ordinary@100B	11,997 natural candidates	2,220 M-missing
Memory action	Ordinary@100B	relation dose plus natural Write $\times$ Use	25M continuations
Paired Use	Ordinary@100B	familiar and independent query surfaces	matched continuations
Context ranking	Ordinary@100B	original and document-disjoint pools	matched continuations
Sequential validation	350M; common initialization	640 common facts; continuous Circuit–Store–Use	5 final routes

The Pythia-style suite (Biderman et al., 2023) complements the target-scale run by isolating functional order under inexpensive same-multiset controls. The selector teacher is instead a custom 350M decoder sharing the primary model’s tokenizer and packed candidate store (Table 6). Transfer consists of example rankings and sequence identities, not model weights or Pythia token IDs. Store and Use continuations start from completed checkpoints; the separate sequential study is specified in Appendix H.

Table 6: Auxiliary architectures and the selector proxy. The Pythia suite and the primary pretraining pipeline have separate tokenizations.

Model	Configuration
Pythia 160M auxiliary	GPT-NeoX; 12 layers; width 768; 12 heads; FFN 3,072; GELU; LayerNorm; official 50,304-token vocabulary and tokenizer
350M selector teacher	349,746,176 parameters; 22 layers; width 1,024; 16-head MHA; SwiGLU 2,816; RMSNorm; RoPE; tied embeddings; primary 65,536-token tokenizer
Teacher training	2,048-token sequences; effective batch 512; AdamW, $\beta = (0.9, 0.95)$, weight decay 0.1; LR 3×10^{-4} to 3×10^{-5} ; 95 warm-up updates; cosine schedule to 3B

B COMMON DIAGNOSIS

B.1 OPERATIONAL RULE

Algorithm 1 Task-conditioned frozen-failure audit

- 1 **Input:** a frozen target, query, candidate sources, checkpoint, and source-retain/remove/value-swap/local-truncation views.
- 2 Assign Circuit when a single unambiguous source exists but binding–matching–transport is unusable.
- 3 Assign Store when no reliable contextual source exists, local context is insufficient, and M-only does not supply the target.
- 4 Assign Use when source content is available but the query-required source or value fails to control the output.
- 5 Preserve Local, mixed, nonactionable, and ambiguous/invalid states; aggregate only actionable excess-loss mass.
- 6 Reapply the same rule at every checkpoint and report frozen-population scope, transitions, and cutoff sensitivity.

For the conflict panel, let p_v be target log probability in view v . Availability is $a = p_{\text{correct only}} - p_{\text{no source}}$; correct-source effect is $c = p_{\text{full}} - p_{\text{mask correct}}$; wrong-source removal gain is $w = \max_j(p_{\text{mask wrong}_j} - p_{\text{full}})$; and local gain is $l = \max(p_{\text{local32}}, p_{\text{local128}}) - p_{\text{full}}$. Availability requires source-only success or a sufficient source-only gain. Actionability requires a remaining full-view failure and causal evidence from history or local context. Ordered rules then identify selection, local–global arbitration, value-use, and binding; unresolved ties remain ambiguous. Table 34 gives the exact gates and precedence.

Rescaling the positive strong-rule thresholds changes few assignments and leaves the actionable class composition nearly unchanged. The moving-residual result is stable across the registered range (Table 10).

B.2 FIXED-CASE TRAJECTORIES

The 32-case panel contains eight cases per retrospective, intervention-refined class. Its labels stay fixed while losses change: Table 7 measures learning within fixed strata, not checkpoint-local reclassification. Local cases retain or improve target support under local32/local128 truncation; Store

cases lack a reliable source and are not explained by that local support. Mixed cases remain separate in the operational audit.

Table 7: Fixed retrospective strata on the same 32 natural cases. C/S/U/L are Circuit/Store/Use/Local excess-mass shares (%); Mass is absolute excess NLL. Each group contains eight fixed identities and retains its label.

Arm	Tokens	C share	S share	U share	L share	Mass
Ordinary	0.3B	25.88	25.35	24.68	24.09	194.42
Ordinary	1.2B	9.80	49.63	2.64	37.92	103.40
Ordinary	3B	4.67	51.95	1.42	41.96	82.27
Ordinary	100B	2.56	44.84	0.00	52.60	60.00
Curriculum	0.3B	25.80	25.00	25.67	23.53	199.02
Curriculum	1.2B	2.96	54.92	2.17	39.95	90.56
Curriculum	3B	4.63	53.08	3.79	38.50	86.82
Curriculum	100B	0.42	47.93	0.03	51.61	53.93

The severity-cutoff sweep preserves the same ordering: Circuit contributes substantially early on but little at the endpoint, where Store and Local account for most remaining mass (Table 10).

Table 8: Selected 942-target conflict cells. Arms: O = Ordinary, C = Curriculum. The C/S/U columns report Circuit/Store/Use shares of actionable excess-NLL mass; Mass is unnormalized.

Arm	Tokens	C	S	U	Mass	Arm	Tokens	C	S	U	Mass
O	0.100	35.74	11.34	52.91	2430.94	C	0.100	33.53	11.87	54.60	2277.94
O	0.481	12.27	10.60	77.12	2132.58	C	0.481	7.78	8.14	84.08	2058.97
O	0.800	5.75	9.39	84.86	1420.16	C	0.800	3.40	10.67	85.93	1209.27
O	1.200	4.74	15.19	80.08	1136.97	C	1.200	1.25	17.21	81.53	926.17
O	3.000	1.99	23.23	74.77	740.32	C	3.000	5.74	23.93	70.33	771.53
O	30.00	2.93	13.39	83.68	536.51	C	30.00	2.35	13.85	83.80	585.74
O	100.0	2.86	15.77	81.37	500.96	C	100.0	1.44	12.02	86.54	532.32

The conflict panel contains 560 H-H and 382 H-L targets and is Use-heavy by construction. Its 100B percentages are conditional residual composition, while the separately frozen natural roster measures case transitions below.

Table 9: Adjudication-free first-checkpoint audit on 272 fixed identities. Mass is absolute excess NLL above 2; solved means full-view target rank = 1.

Arm	Tokens	Circuit cases	Circuit mass	Circuit share	Total mass / solved
Ordinary	0.3B	37	227.10	13.7%	1658.47 / 0
Curriculum	0.3B	50	318.19	19.0%	1673.17 / 0
Ordinary	0.6B	9	41.80	3.1%	1364.17 / 7
Curriculum	0.6B	1	8.14	0.7%	1243.58 / 20
Ordinary	1.2B	0	0.00	0.0%	906.40 / 66
Curriculum	1.2B	0	0.00	0.0%	834.39 / 94
Ordinary	100B	2	5.74	1.0%	596.87 / 103
Curriculum	100B	0	0.00	0.0%	589.97 / 109

For the checkpoint-local audit in Figure 2a, 272 identities are frozen from actionable Ordinary@0.3B behavior and replayed under both arms. No adjudication labels enter the subsequent classification. The enclosing 360-case roster was retrospectively enriched using cross-checkpoint strata, so the audit establishes within-roster dynamics, not natural-token prevalence. Circuit contracts under either Use-first or Local-first precedence; most early Circuit cases are solved by 100B (Tables 9 and 10). These are changes in residual demand, not a requirement that each fact visit all three labels.

Table 10: Residual-audit sensitivity. The threshold-rescaling audit, natural-case severity sweep, and adjudication-free trajectory use their own frozen populations. Percentage shares refer to excess-loss mass.

Audit or quantity	Setting or observed range
Positive strong-rule threshold multipliers	0.5, 0.75, 1.25, 1.5
Changed assignments	0.8–3.1%
Actionable Circuit / Use shares	18.0–19.2% / 80.8–82.0%
Natural-case severity cutoffs	0, 1, 2, 3, 4
Ordinary Circuit share at 0.3B	25.7–26.3% across the cutoff grid
Ordinary Circuit share at 100B	7.1, 4.6, 2.6, 0, 0%, in cutoff order
Endpoint Store + Local share	89–100%
Adjudication-free audit	Thresholds 0–0.5; Use-first / Local-first
Early Circuit mass clears by	1.2B under both precedence orders
Early Circuit cases solved at 100B	Ordinary: 31/37; Curriculum: 43/50

B.3 SAME-FACT WEB-TO-QA COUNTERFACTUAL

The paired panel contains 435 formally reviewed Web failures: 289 semantic questions and 146 cloze items. Each retains its fact, answer span, and target subtoken across presentation views. No-context analysis uses the 241 semantic questions independently judged to identify the same fact without the passage; passage-dependent references and cloze slots are excluded from that view. Aggregate values are reweighted to the frozen Web residual strata used to sample the panel.

Table 11: Same-fact presentation counterfactual. NLL is target NLL; gap is Curriculum-minus-Ordinary. Top-1 reports Ordinary/Curriculum accuracy (%).

View	Cases	Ordinary NLL	Curriculum NLL	Gap	Target top-1
Original Web continuation	435	4.821	5.719	0.898	31.3 / 16.6
Context QA	435	5.288	5.575	0.288	45.1 / 42.1
Paraphrased context QA	435	5.115	5.444	0.329	46.9 / 43.4
Minimal-source QA	435	4.556	4.877	0.321	54.0 / 52.4
Source-masked QA	435	7.300	7.587	0.287	19.3 / 15.6
Shuffled-context QA	435	7.893	8.041	0.148	11.7 / 8.5
No-context QA	241	7.304	7.567	0.263	10.8 / 7.5

The Curriculum–Ordinary gap narrows substantially when the same facts are presented as context QA (Table 11). Relevant evidence helps both arms relative to shuffled context, and masking the correct source removes much of that benefit. Both models therefore use the explicit source; natural Web continuation adds demands of surface realization, source identification, and output selection. Factual/entity and long-tail phrase cases show the largest context-QA reduction, whereas pointer, morphology, and local-relation cases retain more of the original gap.

C CIRCUIT EVIDENCE

CIRCUIT		Build a usable read path	
FIXED	bind → match → transport		ACTION
1	X:A → B ... X:A → [B]		rank during formation joint-to-matching pilot-calibrated exit
2	R:blue ... R → [blue]		
VARY	tokens · domain · source · lexical form		

Figure 7: **Circuit data construction.** Preserve key–match–successor roles while varying content and source realization. Formation-time sensitivity selects natural rows; joint binding–matching followed by matching-only selection orders prerequisites; a pilot calibrates the exit.

C.1 FUNCTIONAL CONSTRUCT

For head h , let $\alpha_h(i, j)$ be attention from position i to j , q the query, v^+ and v^- the correct and decoy value positions, and $(k_r, v_r)_{r=1}^2$ the source pairs. The binding offset $d = v_r - k_r$ is shared by both pairs within a prompt and varies across prompts; $\mathcal{D}$ is the candidate offset set. The clean probe reports binding selectivity A_h , matching selectivity B_h , and value transport C_h :

$$A_h = \frac{1}{2} \sum_{r=1}^2 \alpha_h(v_r, k_r) - \frac{1}{|\mathcal{D} \setminus \{d\}|} \sum_{\delta \in \mathcal{D} \setminus \{d\}} \frac{1}{2} \sum_{r=1}^2 \alpha_h(v_r, v_r - \delta), \quad (7)$$

$$B_h = \alpha_h(q, v^+) - \alpha_h(q, v^-), \quad (8)$$

$$C_h = (W_h^O V_h(v^+) - W_h^O V_h(v^-))^\top (u_{\text{tgt}} - u_{\text{dec}}). \quad (9)$$

Here u_{tgt} and u_{dec} are the unembedding vectors of the correct and decoy tokens. Development and test each contain 128 clean–corrupt pairs of length 192; offset, source distance, slot, query, and token family vary independently. Development selects heads and freezes thresholds; test reports clean–corrupt and component-specific causal gates.

Table 12: Auxiliary 160M functional confirmation. Top: domain-probe binding, matching, and transport controls, where ΔNLL is ablated minus clean. Bottom: natural-carrier confirmation at 100M training tokens in the main and independent runs. Every registered gate passes.

Domain	Binding		Matching			Transport	
	Clean–corrupt	Patch	Clean	Swapped	Q/K ΔNLL	Eligible	V/O ΔNLL
Web	.00703	.00575	.00958	−.11722	.09406	86	.35515
Math	.00748	.00326	.04658	−.15676	.72399	139	.68819
Code	.00690	.00222	.06866	−.15794	1.18290	124	1.14056

Run	Domain	Matching			Binding	Transport
		Clean	Accuracy	Swapped	Ablation drop	Gain
Main 100M	Web	.042	.688	−.163	.045	1.398
Main 100M	Math	.069	.828	−.169	.069	1.763
Main 100M	Code	.098	.836	−.253	.100	2.725
Independent 100M	Web	.041	.758	−.117	.042	1.109
Independent 100M	Math	.048	.852	−.134	.044	1.333
Independent 100M	Code	.093	.766	−.329	.093	2.087

The top panel uses a separate domain-probe pool with 256 clean/corrupt/swapped cases per domain, each of length 512. Binding here is a residual key-readout margin; Patch is its patched-minus-corrupt recovery after inserting clean head activations. Eligible counts cases, not heads: clean binding margin exceeds -0.03191 , matching accuracy exceeds 0.5 , and matching selectivity exceeds 0.0006303 . These are the frozen development gates for this pool, distinct from the 128-pair formal construct above.

Both auxiliary runs reproduce positive clean matching selectivity, negative swap reversal, positive binding-ablation drop, and positive transport gain in every domain (Table 12). The target-scale trajectory runs a fresh dynamic scan at each checkpoint: top-ranked head identities change during formation, with no overlap between the early scan and the endpoint top set. A fixed-reference audit preserves the early Curriculum matching advantage but not the endpoint advantage. Table 35 records the scan and reference checkpoints. Together, these results track formation of a functional path across scales, not one immutable head set.

C.2 TEACHER AND ORGANIZATION

The formation-time and endpoint-time MDA arms select equal-sized, partially overlapping pools (Table 35). Target initialization, optimizer, slots, budget, selected component, and matching-only organization are fixed. MDA uses empirical-Fisher EK-FAC (George et al., 2018) over the prescribed attention Q/K blocks; scores are converted to within-domain percentile ranks before selection.

Table 13: Strict matching-only comparison of formation-time and endpoint-time MDA teachers. Lower is better.

Tokens	Overall NLL		Ambiguous recall NLL	
	Formation	Endpoint	Formation	Endpoint
0.1B	7.58435	7.57876	6.48897	6.46127
0.2B	6.28988	6.30441	4.97248	4.97939
0.3B	5.70661	5.74712	4.37319	4.42689
0.45B	5.10654	5.11048	3.71070	3.66996
0.6B	4.58137	4.59874	2.75410	2.74828
0.9B	4.05616	4.11359	2.00536	2.06120
1.2B	3.79557	3.82161	1.72919	1.74228

Formation-time MDA lowers NLL in every domain at the comparison endpoint, confirming that the formation checkpoint is the stronger teacher for this matching-targeted intervention (Figure 3c).

Table 14: Normalized AUC over 0.1–1.2B for the static teacher-by-organization factorial. Teacher denotes the matching-score checkpoint; the binding-score checkpoint is fixed.

Teacher	Selection order	Binding $\uparrow$	Matching $\uparrow$	Induction $\uparrow$	Overall NLL $\downarrow$	Ambiguous recall NLL $\downarrow$
Formation	Joint-to-matching	.7128	.3071	2.4977	4.8303	3.0933
Formation	Matching-only	.7084	.2964	2.4410	4.8223	3.0960
Endpoint	Joint-to-matching	.6996	.2903	2.4098	4.8392	3.1095
Endpoint	Matching-only	.7141	.2709	2.2298	4.8513	3.1117

Formation-time MDA improves the relevant AUCs within both organizations, and Joint-to-matching improves matching, induction, and ambiguous recall within both teachers (Table 14). An independent update-utility audit also favors the formation teacher before the path forms, then changes sign afterward. Its sample and checkpoint boundary are recorded in Table 35.

C.3 ORDER, EXIT, AND PERSISTENCE

Table 15: Same selected warm-up multiset, different order, evaluated at 2.5B. All columns report PPL except ambiguous recall NLL; lower is better.

Order	Overall	Math	Code	Ambiguous recall NLL	Web
Joint-to-matching	7.6686	18.3830	4.0426	0.6923	41.4245
Interleaved	7.6872	18.4269	4.0561	0.6955	41.3716
Matching-to-joint	7.7018	18.4238	4.0703	0.7026	41.2755

Table 16: Warm-up exit sweep at the common 3B endpoint. Values are PPL; lower is better.

Exit	Overall	Code	STEM	Books	Papers	English Web	Chinese Web
Ordinary	25.9247	3.7900	17.2590	29.6611	21.5048	35.3020	70.5480
0.3B	25.8479	3.7790	17.2387	29.4727	21.5418	35.2636	70.0989
0.6B	25.7401	3.7798	17.2283	29.2783	21.5368	35.1533	69.3255
1.2B	25.6834	3.7703	17.1703	29.2310	21.4604	35.1711	68.9548
2.0B	25.7771	3.7780	17.2160	29.4766	21.4597	35.2412	69.2234

Table 17: Selected validation PPL from the paired 100B run.

Tokens	Ordinary	Curriculum
75B	12.6560	12.6019
80B	12.6177	12.5621
82B	12.6043	12.5497
90B	12.5714	12.5164
95B	12.5625	12.5078
100B	12.5580	12.5034

Every evaluation in the dense late-training window favors Curriculum, as does every endpoint domain (Figure 3d; Tables 17 and 19). The earlier crossing of Ordinary’s final PPL is therefore followed by a sustained advantage. Together with the formation trajectory, teacher comparison, fixed-multiset order test, exit sweep, and independent-initialization comparison, this persistence completes the Circuit evidence chain.

Table 18: Curriculum versus Ordinary across independent 3B initializations. Gains use $100(P_{\text{Ord}} - P_{\text{Curr}})/P_{\text{Ord}}$.

Run	Overall PPL gain (%)	Ambiguous recall NLL direction
1	1.26	lower
2	1.58	lower by 0.00740
3	1.00	higher by 0.00321

Table 19: All six domain PPLs at the matched 100B endpoint.

Domain	Ordinary	Curriculum
Books	15.4227	15.3798
Chinese Web	21.6994	21.3875
Code	2.4922	2.4882
English Web	16.9887	16.9355
Papers	11.2392	11.2190
STEM	9.2482	9.2225

D STORE EVIDENCE

STORE		Separate writing from invocation
FIXED	cue → value	ACTION
1	Who wrote it? → Ada	gate on availability
2	Who authored the notes? → Ada	Memory absent: write
VARY	cue wording · paraphrase · carrier · position	Memory present: invoke

Figure 8: **Store data construction.** Preserve target content and the cue–value relation; vary cue surface, carrier, and optimizer position. Evidence rescue after M-only failure calls for writing; M-only success followed by carrier suppression calls for invocation training.

D.1 GATE AND TIMESCALE

Algorithm 2 M-missing gate and operation-matched action

- Input:** no-source/M-only, evidence-present, and local/template-control views.
- Select M-missing when the no-source candidate decision fails, the evidence-present decision succeeds, and evidence rescue passes the fixed threshold.
- For M-missing, preserve the cue–value relation and vary cues, paraphrases, natural carriers, and optimizer steps.
- If M-only succeeds but both natural carriers suppress its margin, route the case to gated M-USE rather than repeated writing.
- Audit exact relation, strict reverse, unseen paraphrase, wrong pair, non-drag controls, and broad-language cost.

The Store roster is frozen without using future gain as a selection criterion. Its target NLL continues to improve substantially more than that of refined Circuit or Use cases (Table 25). The set overlaps rare n-grams, fixed forms, morphology, entities, facts, and local relations, so prolonged learning is a heterogeneous coverage result rather than a single memory type.

Table 20: Availability-gated and generic hard-example populations, each with 2,048 selected cases. The gated set is frozen from the 2,220 qualifying M-missing candidates. Evidence-present success is part of that selection rule.

Selector	Frozen cases	Evidence-present accuracy
H/L/M M-causal gate	2,048	100.00%
Blind/hard	2,048	31.84%
Random hard	2,048	5.08%

The gate filters natural QA candidates, including TriviaQA (Joshi et al., 2017), by no-source failure and evidence rescue; Table 36 records the retained populations. This operation gate concentrates content-resolvable failures more effectively than generic hard-example selection (Table 20).

D.2 VIEWS AND SELECTION

The natural QA audit compares an original answer with a type-matched replacement. A valid item has distinct answers, a source containing the original answer but not the replacement, and a usable first divergent answer token. At that position, candidate success means $m_\theta > 0$; target-span NLL instead averages all answer tokens. The shared answer prefix is supplied when candidates diverge after the first token.

Table 21: Store input views. The question and candidate answers stay fixed. Evidence availability and answer-free context change separately.

View	Input construction
M-only	Short factual question followed by Answer : ; no source passage.
Evidence	Original answer-containing passage followed by the same question.
Surface	The source sentence truncated immediately before its answer; scored as a completion.
Carrier A / B	The same question preceded by background with answer-containing sentences removed; A uses a Background instruction, B a Notes instruction.

The reference M-missing selector first removes M-only successes, then requires evidence-present success and a margin rescue of at least one logit unit. It ranks eligible items by rescue for fixed-size selection. The surface view diagnoses sentence-prefix effects separately. Blind selection ranks no-source failures by their negative margin; Random hard greedily matches the selected margins using failures outside the M-missing pool.

The reference invocation selector starts from M-only successes. It requires mean drag at least 0.5 and drag at least 0.1 under each carrier, then ranks by mean drag. Controls are M-available facts with mean drag at most 0.1. Greedy matching selects the closest M-only margin among remaining controls of the same entity type and answer-token length, falling back to the remaining pool when that stratum is empty. Controls are used without replacement. Table 36 records the frozen evaluation populations.

D.3 WRITING AND NATURAL USE

The controlled novel-relation panel contains 128 distinct assignments and uses tokenizer-native keys, multiple cues, strict reversed pairs, unseen paraphrases, and wrong-pair training. Packed cue-value segments use block-diagonal attention and independent position IDs; only target spans contribute loss, the target does not appear in the cue, and cue views recur across separate optimizer steps. The total continuation is fixed at 24 optimizer steps; k use M-WRITE and the remainder restore the same ordinary stream.

Table 22: Controlled relation-writing dose. Margin gains are relative to the parent; broad NLL cost is relative to ordinary continuation.

Write steps k	Exact margin	Unseen paraphrase	Broad NLL cost
12	+0.01296	+0.02679	+0.227%
14	+0.03729	+0.02959	+0.298%
15	+0.05009	+0.02910	+0.328%
16	+0.05744	+0.03181	+0.358%
20	+0.07162	+0.03394	+0.674%

The strict-reverse margin exchanges the same two candidates and is the algebraic negative of the exact margin. Unseen paraphrases and the separately trained wrong-pair arm provide the relation controls: correct pairing improves both cue views, whereas wrong pairing favors the swapped assignment (Table 23). The panel thus tests assignment learning on fixed keys and transfer to a held-out cue realization. These controls identify a directional preference for the intended assignment across cue forms. At this dose, the margin response is clearer than candidate-choice accuracy, which does not improve consistently across the two views; the reported result is relation learning measured by margin.

Table 23: Relation controls at the low-dose operating point ($k = 14$). Entries are no-source correct-minus-competing margins; wrong-pair training changes the main-relation margin by -0.0536 while favoring the swapped assignment.

Cue view	Parent	Correct pairing	Wrong pairing
Exact	-0.07528	-0.03799	-0.12884
Unseen paraphrase	-0.03448	-0.00490	-0.04868

For Equation 5, $m_\theta(u; x) = \log p_\theta(y^+ | x) - \log p_\theta(y^- | x)$ compares the correct and competing tokens at their first divergent answer position. Natural-carrier margin is the mean over two fixed carriers. Drag change is continuation-minus-parent margin drag; negative values mean less suppression. The two gain columns instead use parent-minus-continuation target-span NLL, averaged over the same facts (and both carriers for natural use).

For natural use, we select a frozen set from M-available cases with strict circuit drag and cross dragged-versus-control write and use slots. Ordinary rows, tokens, optimizer positions, and continuation budget remain fixed. Table 36 gives the population and slot configuration.

Table 24: Natural Write $\times$ Use factorial. Gains are parent-minus-continuation NLL; larger is better. A/B mean is the natural-use gain in Table 2. Use trains carrier A; B changes the prompt around the same answer-free background and frozen facts. All arms share two intervention slots and a 25M-token continuation, with matched non-drag facts in disabled slots. Parent, ordinary rows, loss, optimizer positions, updates, and budget are fixed.

Arm	Active operation	M-only	Carrier A	Carrier B	A/B mean	Drag change
C0	matched controls	0.636	0.901	0.887	0.894	-0.927
C1	Write	0.740	0.975	0.967	0.971	-0.741
C2	Use	0.723	1.213	1.189	1.201	-1.814
C3	Write + Use	0.806	1.307	1.282	1.294	-1.751

Use contributes the larger gain under each carrier, including the alternative prompt B, while the interaction remains small (Table 24). Matched non-drag controls show why the diagnosis matters: an invocation intervention that helps suppressed facts need not help facts without that failure. Table 25 records these contrasts separately from the dragged-fact factorial.

Table 25: Store audit summaries for separate populations. The first block reports natural-use NLL effects on dragged facts; the second reports drag change on matched non-drag controls (negative means less suppression). The final block reports target-NLL decreases from 3B to 100B on frozen class rosters.

Population	Contrast	Value
Dragged	Write main effect	0.085
Dragged	Use main effect	0.315
Dragged	Write $\times$ Use interaction	0.016
Non-drag control	Write-only drag change	-0.358
Non-drag control	Use-only drag change	+0.393
Frozen Store (Store41)	Target-NLL decrease	1.903
Refined Circuit	Target-NLL decrease	0.297
Refined Use	Target-NLL decrease	0.225
Local	Target-NLL decrease	0.712

E USE EVIDENCE

USE	Apply the downstream source rule	ACTION
FIXED	query + authority → winner	pair opposed winners
1	France: passage says Rome; use passage → Rome	long window:
2	Same passage; use knowledge → Paris	rank full vs local
VARY	fact · query form · source · candidates · winner	

Figure 9: **Use data construction.** In the illustrative conflict, the passage says Rome while established knowledge says Paris. Changing task authority changes the prescribed answer. Preserve that conditional relation across facts and query forms; long-window selection instead concentrates full–local opportunity. Table 26 gives paired, unpaired, and reversed constructions.

E.1 RELATION ENGAGEMENT AND TRANSFER

A worked construction. Table 26 shows the base three-template construction with shortened illustrative facts. Each fact record specifies one question, an edited passage, and two competing answers. A deterministic family hash chooses template t for true pairing; the unpaired arm uses the other two templates and alternates its assigned authority mode across records. Template T_0 asks either to “Use the passage as the authority” or to answer “from established knowledge rather than copying the substitution.” T_1 uses Text/Query/Response fields, and T_2 uses Excerpt/Answer fields, with corresponding passage-authorized and knowledge-authorized instructions.

Table 26: Illustrative paired, unpaired, and reversed training records. C authorizes the passage and M established knowledge. Each listed arrow gives the supervised target; the other answer is its competitor. For readability, both illustrative records are assigned $t = T_0$. The construction follows the implemented allocation; the facts illustrate it rather than report model outputs.

Fact	Question	Edited passage	Knowledge
A	What is the capital of France?	Capital: Rome	Paris
B	What is the capital of Germany?	Capital: Vienna	Berlin
Training arm	Fact A: two decisions		Fact B: two decisions
True paired	$T_0, C \rightarrow \text{Rome}; T_0, M \rightarrow \text{Paris}$		$T_0, C \rightarrow \text{Vienna}; T_0, M \rightarrow \text{Berlin}$
Balanced unpaired	$T_1, C \rightarrow \text{Rome}; T_2, C \rightarrow \text{Rome}$		$T_1, M \rightarrow \text{Berlin}; T_2, M \rightarrow \text{Berlin}$
Reversed	$T_0, C \rightarrow \text{Paris}; T_0, M \rightarrow \text{Rome}$		$T_0, C \rightarrow \text{Berlin}; T_0, M \rightarrow \text{Vienna}$

Both true and unpaired stores contain two decisions per record and equal total counts of the two authority modes; they differ in within-fact mode coverage and template allocation. Ordinary training rows and supplementary loss weights are shared. Familiar evaluation applies all three training templates to unseen families; independent evaluation uses two separately written templates. Training excludes held-out family identities and their original/replacement entities. Thus the familiar-query comparison tests learning the conditional relation across facts, while the independent-query panel tests transfer of that relation across wording.

The positive-control panel fixes 1,000 unseen natural fact families while retaining familiar training query surfaces. True pairing, balanced unpaired carriers, Ordinary, and a reversed relation share the continuation budget and carrier structure.

Table 27: Familiar-query positive control over 1,000 unseen fact families. Accuracy is reported as a percentage; strict family requires every registered surface for a family.

Checkpoint	Arm	Override	Reject	Strict pair	Strict family
+100M	Ordinary	73.57	23.23	4.27	0.10
+100M	Balanced unpaired	73.47	34.70	9.43	1.00
+100M	True paired	73.53	39.67	13.83	1.90
+150M	Ordinary	74.77	22.63	4.47	0.00
+150M	Balanced unpaired	74.10	35.37	10.60	0.80
+150M	True paired	77.13	37.03	14.87	2.90

On an independent matched relation panel, true pairing strengthens the trained route offset and reversing the relation changes its sign (Table 33). This directional response complements the balanced-exposure comparison.

Table 28: Strict-pair accuracy (%) under independently phrased queries. Each repair is evaluated at two saved points.

Round	Ordinary@100M	Paired@100M	Ordinary@150M	Paired@150M
3 templates	7.00	5.45	6.55	4.55
32 templates	2.35	2.95	2.60	2.70
10× families	6.70	2.95	6.60	2.95
Pair-worst	6.70	3.75	6.60	3.85

The independent surfaces show that familiar-query pairing is insufficient for broader transfer (Table 28). They identify query realization and opposed winner direction as dimensions to target in the next conditional-arbitration construction.

E.2 CONTEXT OPPORTUNITY RANKING

The CCUR score is defined in Equation 6. We score successive target blocks using either the full window or their preceding local context; Table 37 gives the window and block sizes. The two truncated views retain identical tokens and either reset positions to zero (L) or retain their original offsets (R). The score averages the clipped minimum loss response over all targets, including zeros. These coordinate conventions implement the same context cut under standard relative-position RoPE; the minimum is the executed scoring rule. LongCE and CCUR match source part, domain, split, full-window-NLL strata, objective, update count, and token budget, isolating the association between full-local response and selected window identity.

Within each source shard, the high-full-NLL half of training rows forms the candidate pool. We stratify by difficulty, rank candidates by CCUR, and select distinct documents under a fixed per-shard quota. LongCE matches each anchor by full-window NLL while excluding that exact window; almost all pairs occupy the same difficulty bin. On the original pool, CCUR beats LongCE at every registered aggregate length and checkpoint. The independent replication excludes every original training document and gives all arms equal exposure to the new pool. Table 37 records the counts and matching configuration.

Table 29: Document-disjoint data-pool replication. All arms share the parent, LongCE objective, matched strata, 92 updates, and 96.47M tokens. Lower answer-token NLL is better.

Checkpoint	Arm	4K	8K	12K
Midpoint	LongCE	4.2271	4.5152	4.4247
Midpoint	Score-shuffled	4.1894	4.4917	4.4060
Midpoint	CCUR	4.1479	4.4368	4.3580
Endpoint	LongCE	3.5081	3.7562	3.8365
Endpoint	Score-shuffled	3.4721	3.7069	3.7906
Endpoint	CCUR	3.4103	3.6264	3.7179

The score-shuffled control preserves the shard/bin score multiset, shard totals, unique-document cap, and budget while breaking the score-to-window relation. CCUR beats both controls at every length and checkpoint (Table 29); paired task-stratified 95% endpoint intervals exclude zero throughout (Table 30). Overall endpoint PPL also favors CCUR (Table 33).

Evaluation uses a fixed synthetic RULER-style (Hsieh et al., 2024) validation set, balanced across word-number retrieval in noise (niah_single_1), retrieval among needles (niah_multikey_2), UUID retrieval (niah_multikey_3), and four-hop variable tracking (vt). We teacher-force the gold answers and microaverage NLL over answer tokens. Case and token counts appear in Table 37. Document disjointness concerns the training pools; all arms share the frozen synthetic evaluation cases.

Table 30: Paired endpoint uncertainty for the document-disjoint training pool. Positive control-minus-CCUR NLL favors CCUR. Intervals resample complete evaluation cases within task, preserving answer-token aggregation (50,000 draws).

Length	Control	NLL gain	95% interval
4K	LongCE	0.0979	[0.0928, 0.1031]
4K	Score-shuffled	0.0618	[0.0574, 0.0663]
8K	LongCE	0.1298	[0.1223, 0.1374]
8K	Score-shuffled	0.0806	[0.0741, 0.0872]
12K	LongCE	0.1186	[0.1109, 0.1264]
12K	Score-shuffled	0.0727	[0.0665, 0.0790]

E.3 CONDITIONAL-ARBITRATION AUDIT

Table 31: Use-localization controls on the original pool. Values are CCUR-minus-LongCE truth-log-probability gains.

View/test	Full	Local only	Full-local	Source removal	Swap winner	Swap value
Gain	.00789	-.00634	.01424	-.00080	-.00490	-.00761

Complete context localizes the average first-pool gain, but source removal and value swaps do not support a semantic winner rule (Table 31). Target-level localization also does not replicate on the document-disjoint models; the repeated positive result is answer likelihood and candidate ranking.

Table 32: Fixed 1,000-family context-versus-memory panel. Accuracy is reported as a percentage; Both requires both query modes for one family.

Checkpoint	Arm	Context override	Memory reject	Both
Midpoint	LongCE	38.55	59.60	1.70
Midpoint	CCUR	38.90	58.65	1.15
Endpoint	LongCE	40.55	55.85	1.30
Endpoint	CCUR	41.10	55.00	1.20

At endpoint, improved context override comes at the expense of memory-authorized decisions; no registered comparison improves both modes and pair completion (Table 32). The domain-matched BridgeSet likewise shows broad gains without Use-specific improvement (Table 33). These controls delimit the positive result as task-aligned window ranking and a source-prior shift, not general H/L/M arbitration.

Table 33: Additional Use diagnostics. Blocks report separate populations and metrics: an independent matched relation panel at +100M, Overall PPL at the document-disjoint endpoint, and a domain-matched 96-case BridgeSet. BridgeSet entries are NLL gains; specificity subtracts the mean gain of the other classes.

Audit	Arm or class	Value
Route offset	Ordinary	0.0282
	True paired	0.1009
	Reversed relation	−0.0884
Overall PPL	LongCE	19.3240
	CCUR	19.1569
BridgeSet NLL gain	Circuit	.00281
	Store	.02539
	Use	.01471
	Local	.01729
	Use specificity	−.00046

F REPRODUCIBILITY

All quantitative figures are generated by `scripts/plot_figures.py` from checked CSV/JSON aggregates under `figures/source_data`. The script emits vector PDFs and inspection PNGs and asserts sample counts, arm/checkpoint identities, mass conservation, and plotted comparison directions. Figure 1, the three construction schematics, and the sequential branch diagram explain task structure or protocol; no image model creates or reconstructs quantitative marks.

The materialized candidate pool is shared across the primary arms (Table 35). Text is normalized, filtered by frozen benchmark substrings, exact-deduplicated, and conservatively near-deduplicated by deterministic LSH bands. The source package includes LaTeX/BibTeX, style files, plotting code, aggregate figure inputs, and vector/PNG figures. It does not distribute corpus text, checkpoints, or every raw per-example array; source licensing, privacy, and redistribution review remain separate from this scientific reproducibility record.

G DETAILED INTERVENTION PROTOCOLS

Conflict-classifier precedence. The classifier first checks validity, availability, and whether a failure is actionable. The nonactionable label means the causal-actionability gate does not fire, rather than requiring a rank-one answer. Strong counterfactual rules then take precedence over frozen adjudication; an ordered fallback handles the remaining cases (Table 34). Recognized adjudications map selection, local competition, use/readout, binding, and parametric coverage to their corresponding fine classes; artifact and unresolved-mechanism judgments map to ambiguous/invalid. Circuit groups binding, Store groups parametric coverage, and Use groups selection, arbitration, and value-use. In this conflict panel, Store enters through recognized parametric-coverage adjudication, not an automated M-only fallback; unadjudicated rows cannot receive Store from these rules. The separate natural-memory gate directly evaluates M-only availability. Only actionable classes contribute excess mass. Missing source-removal effects default to zero.

Table 34: Frozen conflict-classifier rules. Quantities a, c, w, l are defined in Appendix B; r_F and r_L are full and best-local target ranks, and m_F is the full-view margin against the strongest wrong candidate. Within each ordered block, apply the first matching rule. Frozen adjudication uses 360 representative records.

Stage	Condition	Result
Gate	Empty target, replacement character, or no alphanumeric character	Invalid
	$a \geq 0.25$ or source-only target rank = 1	Available source
	$r_F \geq 2$ or full NLL ≥ 2 or $m_F < 0$	Bad output
	Bad output and $[w \geq 0.05$ or $c \geq 0.05$ or $(l \geq 0.10 \wedge r_L < r_F)]$	Actionable
	Valid but nonactionable	Nonactionable
Strong rules, in precedence order		
1	High-confidence frozen artifact judgment	Invalid
2	H–H; available source; $m_F < 0$; $w \geq 0.5$	Selection
3	$m_F < 0$; $w \geq 0.2$; $l \geq 0.5$	Local–global arbitration
4	Available source; $m_F \geq 0$; $r_F \geq 2$; $a \geq 1$	Value-use
5	H–H; unavailable source; $\max(a, c) < 0.25$	Availability/binding
6	$l \geq 1$; $w < 0.2$	Local–global arbitration
7	Recognized high/medium-confidence frozen adjudication	Mapped fine class
Fallback rules, in precedence order		
1	Available source; $m_F < 0$; $w \geq 0.05$	Selection
2	$l \geq 0.10$ and $r_L < r_F$	Arbitration
3	Unavailable source; bad output	Binding
4	Available source; $m_F < 0$; $c < 0.05$	Ambiguous
5	Available source; $m_F \geq 0$; $r_F \geq 2$	Value-use
6	Otherwise	Ambiguous

Teacher and prefix construction. Binding and matching scores use checkpoints appropriate to their formation times; the endpoint teacher supplies the mature comparison. The strict matching-only test holds Q/K components fixed and refits EK-FAC factors at each teacher checkpoint (Table 35). Within each domain, let a_z, b_z be binding and matching score percentiles. Joint binding–matching selection ranks by $\min(a_z, b_z)$ and matching-only selection by b_z . The prefix places joint selection first, then concentrates matching demand, while preserving domain slots and using distinct natural sequences.

Exit calibration and the long run. The target-scale pilot varies the curriculum exit and follows each prefix with ordinary data to a shared endpoint (Table 16). Formation requires consistent binding, matching, and read behavior at adjacent checkpoints, positive ambiguous recall loss under

dynamically located head ablation, persistence into ordinary continuation, and held-out Overall/Web performance. The best validation exit occurs after the earliest passing formation checkpoint. The long run fixes that schedule in advance, then follows ordinary training through 100B tokens. The pilot therefore supplies a measured formation window, while the long run tests a frozen schedule derived from it; exact operating points are in Table 35.

Table 35: Circuit data and protocol settings. These are fixed experimental instances of the construction described in the text; teacher checkpoints and target-model checkpoints are distinct.

Item	Setting
Teacher parameters	349,746,176
Teacher checkpoints	Binding: 200.278M tokens; matching formation: 1.501B; endpoint: 3.000B
Matching components	Teacher Q/K heads L9H5 and L9H13
Prefix allocation	Joint selection 40%; matching-only 60%
Primary prefix	234,701 joint + 352,051 matching-selected sequences; 1.2017B tokens
Domain slots	Code 10%, STEM 10%, Books 25%, Papers 15%, English Web 25%, Chinese Web 15%
Validation	37,718,637 shared target positions
Materialized candidate pool	60,951,155 packed sequences; 124,827,965,440 target tokens
Teacher-time selected pools	585,728 sequences/arm; 280,560 shared; Jaccard 0.3149
Dynamic head audit	No 0.3B head remains in the endpoint top-eight set
Fixed-reference audit	Ordinary@0.8B; early advantage through 1.2B, absent at 100B
Matching update-utility audit	384 natural sequences; formation-minus-endpoint utility positive through 0.6B, then changes sign
Pilot exits / endpoint	0.3, 0.6, 1.2, 2B / 3B tokens
Persistence requirement	At least 0.5B ordinary tokens after the candidate exit
Earliest passing exit	0.6B
Frozen long-run prefix	1.2B; best pilot Overall PPL at 3B
Dense late-training audit	16 evaluations from 80B through 100B

Memory treatment and control slots. The natural factorial pairs dragged facts with matched non-drag controls. Each arm repeats fixed Write and Use slots at the same optimizer positions (Table 36). Active slots draw dragged facts; disabled slots draw matched non-drag facts in the same format. Write presents the cue–value relation, while Use presents its natural carrier. Independently attended segments each carry one supervised decision target. The arm definitions in Table 24 specify which operation is active; remaining updates use the same ordinary rows. All arms resume the same model and optimizer. Training and evaluation share the frozen facts. Write uses the sentence prefix ending at the answer; Use trains carrier A, while evaluation reports M-only and both A/B prompts around the same answer-free background. The intervention loss marks only the first divergent answer token, whereas the reported NLL gains average the target span. The controlled novel-relation study separately holds out a cue realization while retaining the learned assignments.

Paired route objective and evaluation units. A training pair contains context-authorized and memory-authorized decisions for one fact family (Table 26). Balanced unpaired supervision uses the same fact records and two decisions per record, assigns only one mode to each record, and balances modes across records. It uses the other two query templates; true pairing uses both modes of one template. Reversed supervision keeps the true-paired inputs and exchanges the target and competitor labels at the decision positions. Ordinary next-token loss is supplemented by $\lambda[\text{softplus}(z^- - z^+) - \log p_\theta(y^+ | x)]$, where z^+, z^- are target and competitor logits at the first divergent answer token and p_θ is the full-vocabulary softmax. The supplementary loss is additive over decisions. Table 36 gives λ and the evaluation units. A decision is correct when the authorized candidate wins at its first divergent token. Strict pair requires both authority modes for a row; strict family requires all registered surfaces. With authorized-minus-competing margins m_C and m_M for

context and memory modes, route offset is $\mathbb{E}[(m_C + m_M)/2]$. It measures signed relation alignment, while strict-pair accuracy requires both decisions to succeed. Independent phrasings change the query surface, while the additional controls vary template diversity, family coverage, and emphasis on the less successful direction (Table 28). The final saved checkpoint follows an ordinary-training retention stage.

Table 36: Store and paired-Use protocol settings. Gate candidates, natural invocation cases, and route-evaluation families are separate populations.

Item	Setting
M-missing gate	12,000 candidates; 11,997 valid; 2,220 M-missing
Natural-use filtering	11,468 complete-view candidates; 7,247 M-available; 2,212 strict-drag cases
Natural factorial	512 dragged facts + 512 matched non-drag controls
Slot schedule	Write and Use slots, each repeated 3 times among 24 updates
Packed row	Four independently attended 512-token segments; one supervised target/segment
Parent / learning rate	Same 100B model and optimizer; 2×10^{-5}
Paired-route training and evaluation	
Base relation store	8,192 rows; 16,384 decision targets; 8,192 per authority mode
Relation schedule	Every second update during the 96-update relation stage
Objective coefficient	$\lambda = 0.00075$ for each supplementary term
Familiar-query evaluation	1,000 unseen families; 3 surfaces/family; 3,000 rows
Independent-query evaluation	2 surfaces/family; 2,000 rows
Retention checkpoint	+100M relation stage, then 50M ordinary tokens; final +150M

Long-context objective and extension. All long-context arms extend the same 2K parent to 16K and resume under matched optimization (Table 37). LongCE (Fang et al., 2025) uses full-context next-token loss with detached weights $w_i = \min\{\exp(\ell_i^L - \ell_i^F), 5\}$ on scored positions and unit weights on the unscored local-history prefix. The loss is the mean of $w_i \ell_i^F$ over target tokens. The local teacher uses the selector’s same history/block construction. The training weight uses the reset-position local view L ; the original-position view R supplies the additional guard in the selection score, not a second training weight. LongCE, CCUR, and score-shuffled arms share this objective and context extension; selection changes their complete training windows.

Table 37: Long-context scoring, selection, and evaluation settings. The independent pool excludes all original training documents. All arms within a pool share the stated configuration.

Item	Setting
Full window / scored positions	16,384 tokens / 4,096–16,383 (12, 288 targets)
Target blocks / local history	Twelve 1,024-target blocks / 4,096 preceding tokens
Context extension	2K→16K; linear RoPE scaling by 8 (Chen et al., 2023b)
Learning rate / batch	2×10^{-5} / 64 complete windows per update
Original window pool	126,170 windows across 8 source shards
Difficulty strata	High-full-NLL half of each shard; 20 equal-frequency bins
Original selection	768 distinct-document windows/shard; 6,144 total
LongCE anchor matches	6,105 same-bin pairs; 39 adjacent-bin pairs
Original training budget	100.66M tokens per arm
Independent pool exclusion	All 10,542 original documents
Independent selection	184 windows/shard; 1,472 total; 4 equal exposures
Independent continuation	92 updates; 96.47M tokens
Evaluation lengths	4K, 8K, 12K
Evaluation cases	800/length; 200/task
Answer tokens by length	10,597 / 10,594 / 10,668

H SEQUENTIAL VALIDATION

H.1 ONE CONTINUOUS TRAINING CHAIN

The sequential study uses an approximately 350M-parameter architecture and one tokenizer. Routes A (ordinary order) and B (early Circuit curriculum) start from the same initialization. Their pretraining data multiset, per-position domain allocation, optimizer, and learning-rate schedule are matched. They are parallel routes: B does not inherit the teacher’s model parameters. B uses 458 Joint updates, 686 Matching updates, and 1,717 ordinary updates; A uses ordinary order throughout. This prefix schedule is fixed before the common-task endpoints are evaluated.

Store inherits its route’s model, optimizer, and random state. Use inherits the corresponding Store state; branches with the same parent start from the same saved state. Figure 10 shows the four Store parents and five final routes. Fixed schedules test composition of the interventions, rather than online switching between them.

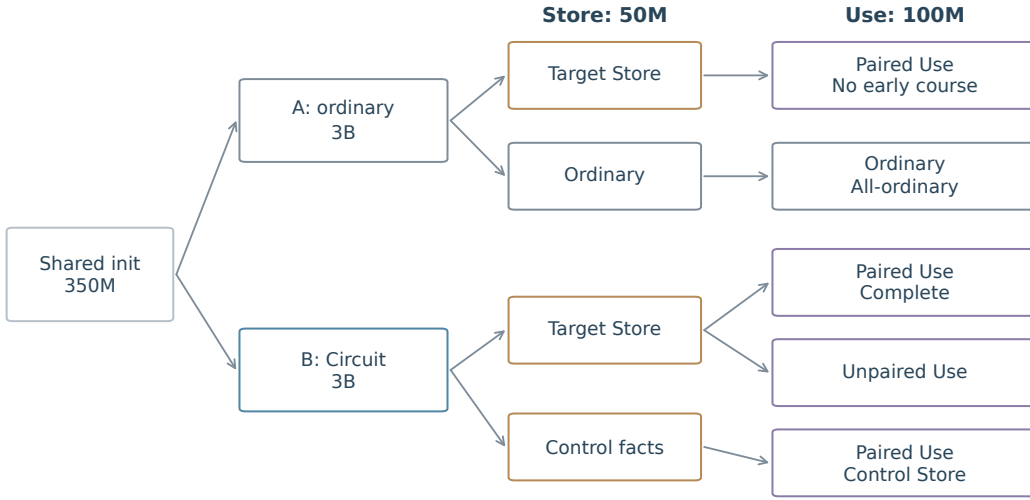


Figure 10: **Continuous branches with a shared initialization.** A/B differ in early pretraining order. Target and control Store write matched but different facts. The paired/unpaired comparison shares a target-Store parent. Only arrows denote model-state inheritance.

Table 38: Sequential training settings. Positions are model-input positions, not parameter counts or GPU costs.

Stage	Updates	Effective batch	Input positions	Learning rate
Pretraining (each A/B)	2,861	512	2,999,975,936	shared schedule
Store (each parent)	381	64	49,938,432	3×10^{-5}
Use (ordinary stream)	96	512	100,663,296	2×10^{-5}

All ordinary sequences have 2,048 positions. Store keeps stage budget, ordinary-text schedule, teaching slots, and exposures per fact fixed. Use keeps ordinary inputs, auxiliary-input count, teaching facts, and effective updates fixed. The ordinary-continuation control receives the same auxiliary forward inputs with zero relation weight; this matches input volume, not auxiliary-backward compute. The two trunks, four Store parents, and five Use endpoints total approximately 6.71B input positions including auxiliary inputs.

H.2 FACTS, SUPERVISION, AND EVALUATION

The target pool contains 1,280 facts selected using A’s pretraining endpoint: the model fails the no-evidence candidate decision and succeeds when correct evidence is supplied. Another 1,280 real facts provide matched controls by answer type, text length, and format. Target membership is not reselected using B or continuation outcomes. The comparison is therefore on this A-selected common pool.

Table 39: Fixed target-fact split. The main sequential results use only the 640 common tracking facts.

Partition	Facts	Store-taught	Use-taught
Use teaching	384	yes	yes
Common tracking	640	yes	no
Development	256	yes	no

Target Store writes the target questions and answers; control Store uses matched control questions and answers. Each fact receives 19 teaching exposures at distinct updates. A sequence contains four independently attended QA segments, with supervision at the first token where the correct and competing answers differ. Teaching and ordinary updates follow a fixed interleaving schedule.

For Use, each teaching fact has a true answer and a passage with a conflicting answer. One instruction authorizes the passage; the other authorizes established knowledge. Paired and balanced-unpaired constructions each contain 384 facts, 384 relation rows, and 768 decisions, balanced across the two authority modes. Paired teaching gives both modes for one fact and template; unpaired teaching gives one mode per fact using the other two base templates. Thus fact count, decision count, overall mode balance, and loss are matched, while within-fact authority coverage and template allocation differ.

Use adds target cross-entropy and target-versus-competitor ranking losses, each weighted by 0.00075, to ordinary language modeling. Every second ordinary update includes eight relation rows (16 decisions), for 48 auxiliary batches. Ordinary and auxiliary gradients enter the same optimizer update, with accumulation preserving the effective batch.

Every node is evaluated on identical common facts and queries: one correct-evidence reading question per fact, four no-evidence memory wordings per fact, and three familiar authority templates with two decisions each. The latter gives 1,920 pairs and 3,840 decisions. Repeated templates are not additional independent facts. Answer NLL sums loss over all tokens of the full correct answer and divides by the total answer-token count in that view. Joint-use NLL pools both authority modes, so a lower average does not imply improvement in each mode. Pair accuracy requires the authorized candidate to win at its first divergent token under both instructions; it is not complete-answer generation accuracy.

H.3 STAGE AND ENDPOINT RESULTS

Table 40: All 18 stage-trajectory values on the same 640 common facts. A uses ordinary pretraining; B uses the early Circuit curriculum. Both then receive target Store and paired Use. Lower answer-token NLL is better.

Route	Node	Reading NLL	Memory NLL	Joint-use NLL
A	Pretraining	3.156392	6.025372	3.863549
A	Store	2.179583	4.426465	3.192157
A	Use	2.170958	4.371545	3.050034
B	Pretraining	3.136392	5.995372	3.839549
B	Store	2.139583	4.346465	3.132157
B	Use	2.130958	4.291545	2.970034

Table 41: Within-stage NLL reductions on the common facts. The final column is the difference of the two observed improvements, not a fitted interaction or significance estimate.

Contrast	A	B	B minus A
Store: memory before minus after	1.599	1.649	0.050
Use: joint-use before minus after	0.142	0.162	0.020

Table 42: All five final routes on the common task. Pair-NLL win rate sums the two full-answer NLLs within a query pair and counts when the complete route is lower than the listed control; ties are not wins. Its denominator is 1,920 pairs, and it is distinct from candidate accuracy.

Route	Memory NLL	Joint-use NLL	Pair accuracy	Pair-NLL wins
Complete sequence	4.292	2.970	5.26%	—
No early curriculum	4.372	3.050	3.39%	58.28%
Control facts in Store	5.220	3.245	1.41%	79.74%
Unpaired Use	4.289	3.019	3.18%	55.10%
Ordinary continuation	6.026	3.870	0.99%	91.77%

The complete sequence improves the common use task over every replacement, including no early curriculum. Unpaired teaching has slightly lower memory NLL but weaker joint use, separating answer availability from task-authorized selection. The absolute pair accuracy remains low; the positive result is the sequential improvement under familiar instructions, consistent with the broader arbitration frontier in Section 6.

H.4 A COMMON-FACT HANDOFF

The worked fact asks when Newton died. Established knowledge gives 1727; the fixed conflict passage reads: “Newton died in 1790. Poems honoring his scientific work continued to appear for decades after his death.” Passage-authorized queries require 1790; knowledge-authorized queries require 1727. This fact is Store-taught and Use-untaught. Table 43 follows the same fixed question and candidate pair at each node. It is an explanatory case, not an estimate of how often this sequence of errors occurs.

Table 43: One fact across the continuous chain and controls. Margins are authorized-minus-competing first-divergent-token scores in the reading and no-evidence views. Pair NLL sums the two authority decisions for the fixed query pair.

Node or control	Reading margin	Memory margin	Pair NLL	Both correct
A pretraining	+1.375	−0.469	5.144	no
B pretraining	+1.450	−0.375	5.100	no
B + target Store	+6.250	+8.938	3.300	no
Complete sequence	+5.062	+7.031	2.850	yes
No early curriculum	+4.188	+5.312	3.011	no
Unpaired Use	+4.312	+6.469	2.975	no

Reading already succeeds before Store, but unaided recall does not. Store establishes the correct memory candidate without resolving the conflict pair; subsequent paired Use resolves that pair. This illustrates the distinction between a usable read path, available content, and query-conditioned control without requiring every fact to follow an identical trajectory.